Analysis of Influencing Factors of Consumer Confidence Index: An Empirical Study

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Abstract

This study aims to analyze the key factors influencing the Consumer Confidence Index (CCI) and evaluate their effectiveness in predicting CCI. The research first employs Pearson correlation coefficients, Lasso regression, and Recursive Feature Elimination (RFE) to screen a wide range of potential factors and identify key features. Subsequently, Linear Regression, Decision Tree, and Random Forest models are trained to examine their predictive performance. Furthermore, feature importance scores from Decision Tree and Random Forest models are used to assess the contribution of each variable to CCI. Finally, using key features, deep learning models such as Long Short-Term Memory (LSTM) and Multi-Layer Perceptron (MLP) are applied as benchmark models to analyze their performance in forecasting the consumer confidence index. The results provide empirical evidence for understanding the drivers and prediction of consumer confidence.

Keywords: Consumer Confidence Index; Machine Learning Models; Empirical Study

1 Introduction

The Consumer Confidence Index (CCI) is a vital indicator for measuring consumers' views on current economic conditions and future expectations, playing a key role in economic research and policy-making. High consumer confidence usually signals an increase in consumer spending, which is a primary driver of economic growth. Therefore, understanding the factors that influence consumer confidence is of great significance for promoting economic stability.

In recent years, especially after the COVID-19 pandemic, China's CCI has remained consistently below 100. This reflects consumers' concerns and uncertainty about economic prospects. Despite government stimulus measures, recovery has been slow due to long-term economic impacts, employment pressure, and income instability. According to the National Bureau of Statistics, since 2023, China's CCI has frequently fallen below historical averages. This low confidence inhibits demand and poses risks of deflation. Boosting consumer confidence has become a core goal for China's high-quality development.

Scholars have used various methods to study CCI. Sun (2021) used web search indices and news text with machine learning (Lasso, MLP) to predict CCI. Tang (2020) combined web search data with LSTM for multi-term predictions. Wang (2020) used Vector Autoregression (VAR) to find that CPI changes significantly affect CCI with a time lag. Other researchers, such as Chen (2020), analyzed the relationship between cultural industry upgrades and CCI. This study aims to build upon these works by using a comprehensive machine learning framework to identify and validate the most significant drivers of China's CCI.

2 Methodology and Data

2.1 General Framework

Our research logic begins with feature engineering to select relevant variables, followed by feature importance ranking using tree-based models, and finally utilizing deep learning to capture non-linear relationships.

1. **Feature Selection:**
   - *Pearson Correlation:* Preliminary screening of highly correlated variables.
   - *Lasso Regression:* Using L1 regularization to remove non-significant features.
   - *Recursive Feature Elimination (RFE):* Iteratively removing the least important features to find the optimal subset.
2. **Model Training and Prediction:** Testing Linear Regression (baseline), Decision Tree (non-linear), and Random Forest (ensemble stability).
3. **Feature Importance Analysis:** Evaluating variable contributions through model scores to identify key drivers.
4. **Deep Learning Application:** Utilizing LSTM for time-series features and MLP for non-linear mapping.

2.2 Data Sources and Preprocessing

Data was collected using the `Akshare` API, covering macro-economic indicators from January 2016 to September 2024.

- **Indicators:** GDP, Unemployment, CPI, PMI, SSE 50 Index, Import/Export, Fixed Asset Investment, and Retail Sales.
- **Preprocessing:** Missing values were handled, and time-series alignment was performed. Since GDP is quarterly, forward-filling was used. Stock data was averaged by month. Difference processing (growth rates) was applied to ensure stationarity, and all features were normalized.

2.3 Descriptive Statistics

Table 1 summarizes the main variables used in the study.

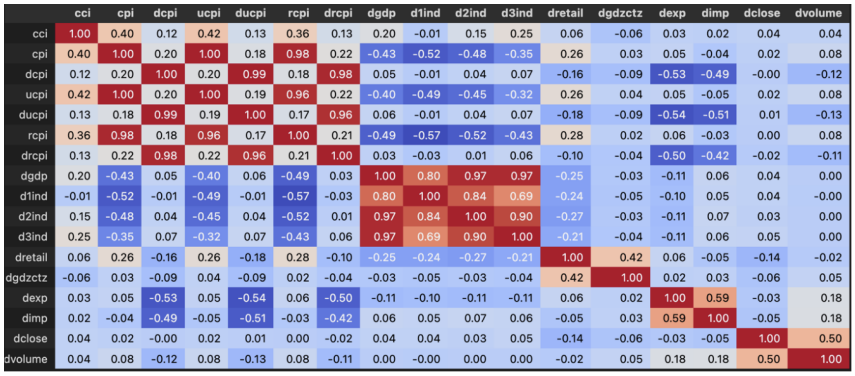
Table 1: Descriptive Statistics

Variable	Count	Mean	Std	Min	25%	50%	75%	Max
cci	106	109.2	85.5	89.2	116.1	122.2	127.6	15.1
dcci	106	-0.1	-23.4	-1.1	0.2	1.4	4.7	3.2
csi	106	105.3	84.2	88.9	110.6	117.2	121.0	13.3
cpi	106	101.6	99.2	100.7	101.8	102.3	105.4	1.2
dgdp	105	5.6	-6.9	4.8	6.1	6.8	18.3	3.9
d1ind	105	3.7	-3.1	3.1	3.5	4.1	8.1	1.9
d2ind	105	5.5	-9.7	4.3	5.8	6.1	24.4	4.9
d3ind	105	6.0	-5.4	4.6	7.2	7.7	15.6	3.7
retail	106	34107.6	24645	30415	34237	37913	45396	4862.6
dgdzctz	106	-2.7	-65.2	-10.1	-3.5	6.5	32.1	18.3
dexp	106	2.0	-62.0	-2.4	1.8	6.2	129.6	19.5
close	106	3133.0	2558.4	2949.6	3135.1	3292.5	3627.2	247.2

3 Empirical Modeling and Analysis

3.1 Feature Selection

To select factors influencing CCI, we first used Pearson Correlation (Fig 1). The top 8 features were identified through various methods (RFE, Lasso). By taking the union, we selected 15 initial variables for the model.



	cci	cpi	dcpi	ucpi	ducpi	rcpi	drcpi	dgdp	d1ind	d2ind	d3ind	dretail	dgdzctz	dexp	dimp	dclose	dvolume
cci	1.00	0.40	0.12	0.42	0.13	0.36	0.13	0.20	-0.01	0.15	0.25	0.06	-0.06	0.03	0.02	0.04	0.04
cpi	0.40	1.00	0.20	1.00	0.18	0.98	0.22	-0.43	-0.52	-0.48	-0.35	0.26	0.03	0.05	-0.04	0.02	0.08
dcpi	0.12	0.20	1.00	0.20	0.99	0.18	0.98	0.05	-0.01	0.04	0.07	-0.16	-0.09	-0.53	-0.49	-0.00	-0.12
ucpi	0.42	1.00	0.20	1.00	0.19	0.96	0.22	-0.40	-0.49	-0.45	-0.32	0.26	0.04	0.05	-0.05	0.02	0.08
ducpi	0.13	0.18	0.99	0.19	1.00	0.17	0.96	0.06	-0.01	0.04	0.07	-0.18	-0.09	-0.54	-0.51	0.01	-0.13
rcpi	0.36	0.98	0.18	0.96	0.17	1.00	0.21	-0.49	-0.57	-0.52	-0.43	0.28	0.02	0.06	-0.03	0.00	0.08
drcpi	0.13	0.22	0.98	0.22	0.96	0.21	1.00	0.03	-0.03	0.01	0.06	-0.10	-0.04	-0.50	-0.42	-0.02	-0.11
dgdp	0.20	-0.43	0.05	-0.40	0.06	-0.49	0.03	1.00	0.80	0.97	0.97	-0.25	-0.03	-0.11	0.06	0.04	0.00
d1ind	-0.01	-0.52	-0.01	-0.49	-0.01	-0.57	-0.03	0.80	1.00	0.84	0.69	-0.24	-0.05	-0.10	0.05	0.04	-0.00
d2ind	0.15	-0.48	0.04	-0.45	0.04	-0.52	0.01	0.97	0.84	1.00	0.90	-0.27	-0.03	-0.11	0.07	0.03	0.00
d3ind	0.25	-0.35	0.07	-0.32	0.07	-0.43	0.06	0.97	0.69	0.90	1.00	-0.21	-0.04	-0.11	0.06	0.05	0.00
dretail	0.06	0.26	-0.16	0.26	-0.18	0.28	-0.10	-0.25	-0.24	-0.27	-0.21	1.00	0.42	0.06	-0.05	-0.14	-0.02
dgdzctz	-0.06	0.03	-0.09	0.04	-0.09	0.02	-0.04	-0.03	-0.05	-0.03	-0.04	0.42	1.00	0.02	0.03	-0.06	0.05
dexp	0.03	0.05	-0.53	0.05	-0.54	0.06	-0.50	-0.11	-0.10	-0.11	-0.11	0.06	0.02	1.00	0.59	-0.03	0.18
dimp	0.02	-0.04	-0.49	-0.05	-0.51	-0.03	-0.42	0.06	0.05	0.07	0.06	-0.05	0.03	0.59	1.00	-0.05	0.18
dclose	0.04	0.02	-0.00	0.02	0.01	0.00	-0.02	0.04	0.04	0.03	0.05	-0.14	-0.06	-0.03	-0.05	1.00	0.50
dvolume	0.04	0.08	-0.12	0.08	-0.13	0.08	-0.11	0.00	-0.00	0.00	0.00	-0.02	0.05	0.18	0.18	0.50	1.00

Figure 1: Correlation Coefficient Heatmap

3.2 Model Training Results

We used Mean Squared Error (MSE) and R-squared (R^2) to evaluate models. Results in Table 2 show that Tree-based models significantly outperform linear ones, indicating non-linear relationships.

Table 2: Model Training Results

Category	Model Name	MSE	R-squared	Rank
Machine Learning	Decision Tree	15.03	0.93	1
	Random Forest	34.78	0.83	2
	OLS Regression	220.11	-0.09	4
	KNN Regression	236.47	-0.17	5
	SVM Regression	243.58	-0.21	6
Deep Learning	LSTM	103.49	-52.06	3
	MLP	449.24	-1.23	7

3.3 Feature Importance Analysis

Based on Decision Tree and Random Forest scores, Table 3 ranks the importance of factors. GDP growth rates and overall GDP are the top contributors.

Table 3: Feature Importance Scores

Feature	Rank	Decision Tree	Random Forest	Average
d3ind	1	0.485268	0.235364	0.360316
d1ind	2	0.396715	0.151010	0.273863
dgdp	3	0.084826	0.344436	0.214631
d2ind	4	0.025606	0.093720	0.059663
dretail	5	0.001323	0.029376	0.015350
cpi	6	0	0.025271	0.012636

4 Conclusion and Policy Recommendations

This study identifies CPI, GDP, and industrial growth as core factors influencing consumer confidence. Tree-based models provide the highest predictive accuracy for CCI.

Policy Suggestions: 1. Prioritize GDP growth, especially in the service sector. 2. Maintain stable inflation (CPI) to ensure market liquidity. 3. Utilize machine learning for real-time monitoring of consumer sentiment.

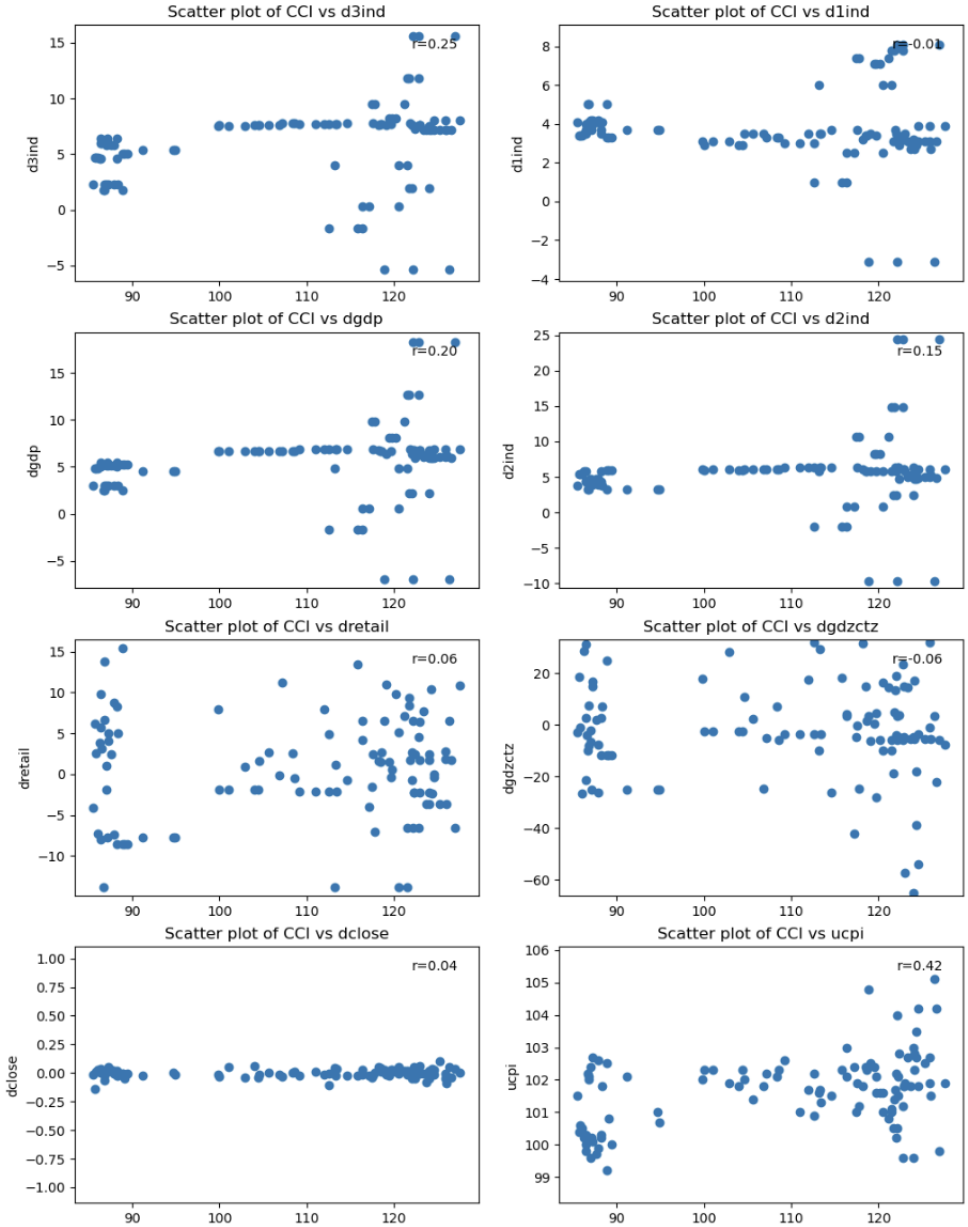


Figure 2: Scatter Plots of CCI vs. Key Factors

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